

RESEARCH

Intelligent Recommendation of Information Resources in University Libraries Based on Fuzzy Logic and Deep Learning

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Abstract

University libraries contain vast information, yet students often struggle to locate relevant resources efficiently to support their academic learning needs. Traditional search and recommendation systems suffer from low personalization and limited handling of vague or uncertain user preferences, reducing their effectiveness in complex educational environments. To address these limitations, this research proposes the Fuzzy Deep Learning-Based Intelligent Library Resource Recommendation Framework (FDLILRRF), which integrates fuzzy logic and deep learning techniques to enhance the accuracy and relevance of information retrieval in university libraries. The research focuses on developing smart systems to retrieve information and recommend educational resources tailored to individual needs. The core problem identified is the inadequacy of keyword-based search engines to deliver personalized and context-aware resource suggestions within large digital library systems. The FDLILRRF framework uses fuzzy logic to handle unclear user input and deep learning to learn from interactions and content, ensuring personalized recommendations. This hybrid approach enables the generation of more accurate and context-sensitive recommendations. Potential applications of this framework include university digital libraries, online learning platforms, and research support systems, offering significant benefits in academic resource discovery and user satisfaction. Experimental evaluation using a student interaction dataset demonstrated that FDLILRRF improved recommendation accuracy by 14.6% over traditional collaborative filtering methods. Enhanced performance in precision and recall metrics further confirms the framework's effectiveness for real-world educational applications. This research provides an intelligent, adaptive solution to optimize information access in academic library systems.

Keywords Intelligent recommendation system · University library resources · Fuzzy logic · Deep learning · Educational information retrieval · User interaction analysis

1 Introduction

University libraries are vital knowledge sources for research, teaching, and learning across disciplines [1]. Modern academic institutions contain millions of books, journals, research papers, theses, multimedia, and specialist databases. Scholarly knowledge discovery and utilization are hampered by information overload produced by 2.5 million scholarly papers per year [2]. Keyword search techniques and human classification are used for university library information retrieval. Traditional systems need specific queries using specialized vocabulary, which contradicts with



researchers' and students' natural language patterns and conceptual thinking [3]. Traditional libraries struggle to adapt to user preferences, learning habits, and shifting research interests, resulting in unsatisfactory resource discovery [4].

The digital transformation of university libraries has revolutionized information management and user interaction. Digitization has increased accessibility and storage, but it has also made locating essential information in massive archives more challenging [5]. It might take time and impede scholars from locating relevant academic information when they struggle to browse vast databases. The research examines the inefficiency of university library information retrieval systems. Traditional systems average 45–55% precision and 2.8–3.2 user satisfaction [6]. Three major technical issues are the semantic gap, the personalization deficit, and the scalability.

The semantic gap occurs when user queries contain ambiguous or imprecise language, whereas the personalization deficit is due to static algorithms that do not consider individual user preferences or contextual factors [7]. The scalability challenge arises from the computational complexity of processing large-scale library datasets [8].

Current university library recommendation system technologies are lacking, according to 127 peer-reviewed studies published between 2018 and 2024. Due to the scarcity of user–item interactions in library systems, collaborative filtering algorithms that work in commercial e-commerce settings perform poorly [9]. Content-based filtering enhances performance but over-specializes, leading to restricted recommendations and low diversity. Hybrid recommendation systems yield increases of 3–7% over single-method approaches due to integration issues [10]. Weak machine learning techniques in library recommendation systems fail to capture deep semantic links between resources and user intents. Only 12% of reviewed library science papers used neural network designs [11]. Other research gaps include inflexible matching algorithms that are unable to process natural language queries with uncertainty and ambiguity. Fuzzy logic has been neglected in library recommendation system design [12].

An intelligent recommendation system for university libraries is essential for academic research and study. The exponential growth of digital library collections necessitates the development of improved algorithms to process high-dimensional data with acceptable response times [13]. The semantic comprehension issue requires advanced natural language processing to bridge the gap between human conceptual thinking and machine-readable metadata [14]. Optimizing user experience requires efficient data structures, optimal algorithms, and scalable system architectures for real-time personalization. Managing the strengths and shortcomings of numerous recommendation methods is an integration challenge [15]. To address these limitations, this research proposes the Fuzzy Deep Learning-Based Intelligent Library Resource Recommendation Framework (FDLILRRF). The main objectives are:

- To design a hybrid recommendation system that integrates fuzzy logic, deep neural networks (DNN), and content-based filtering for personalized information retrieval in university libraries.
- To handle uncertainty and vagueness in user input using fuzzy logic for more accurate and human-like interpretation of queries.
- To improve recommendation accuracy and relevance by training a deep neural network on user behavior and library resource features.
- To evaluate the effectiveness of the proposed system using real-world library datasets and standard metrics, such as accuracy, precision, recall, and mean reciprocal rank (MRR).

A list of the research monitors. The second part presents a comprehensive literature review and outlines the research methods. Section 3 discusses the research plan, methods used, and data processing. Section 4 talks about what the analysis found. Section 5 has the conclusion and plans for the future.

2 Related Works

2.1 Fuzzy Logic Applications in Library Information Systems

Verma and Patnaik [16] suggested a ranking-based hybrid recommendation system for college libraries to help students choose from many books. Preprocessing labels department names, weights books by issue and return

dates, and computes the Pearson correlation coefficient (PCC) to discover department similarities. By ranking books, the Hidden Markov-assisted Chaotic Artificial Humming Bird with Discriminant Analysis and weighted fuzzy ranking (HMCAHB_DA-WFR) model improves suggestions. The approach utilizes Python and a real-time dataset from the Bhilai Institute of Technology, Durg. Better accuracy, Kappa, MAE, NMCC, FPR, and FNR. The simulation results demonstrate that the proposed strategy outperforms other strategies.

Oughannou et al. [17] explored the use of IoT and fuzzy logic control in education, particularly to optimize classroom temperature comfort. It highlights how ICT enhances education and how IoT is utilized in various educational settings. The innovative classroom architecture utilizes Fog Computing and Edge Computing to address storage, connectivity, and real-time analysis requirements. The article also discusses MQTT architecture and its role in client–server computing. The study highlights how the integration of IoT and fuzzy logic control enhances thermal comfort, revolutionizing classroom management and optimization. The integration of technology into universities will provide lasting and sensible solutions to current problems, according to the paper.

2.2 Deep Learning Techniques for Resource Recommendation in Digital Libraries

Sa'ari et al. [18] provided that integrating deep learning (DL) algorithms into libraries is complicated by ethical issues, a lack of specialist knowledge, dataset selection, human interaction, and cognitive constraints in the brain. A rigorous literature review is necessary to comprehend digital literacy in libraries. This study utilizes deep learning (DL) algorithms to analyze user-seeking behavior, aiming to enhance user experience and personalize services. A thorough search and screening are done using Scopus. DL algorithms improve information retrieval and user-centric services by personalizing recommendations, resource suggestions, and search results. To optimize the library DL algorithm's potential, ethics and research are emphasized. Libraries benefit from the improved information retrieval, resource recommendation, and user-centric services offered by DL algorithms.

Liu [19] investigated sophisticated recommendation systems utilizing deep learning algorithms in the university library. This study examined 5 years of library data, including book records, borrowing transactions, and related information. College libraries foster talent and drive innovation in library services, according to the research. The paper critiques book recommendation systems and suggests a new service architecture. For the best book recommendation for teachers and students, the artificial intelligence recommendation system is used. The paper demonstrates how deep learning algorithms can enhance library services and facilitate the onboarding of new users.

2.3 Hybrid Recommender Systems Combining Fuzzy Logic and Deep Learning

Kumar et al. [20] have given the amount of pedagogical content, Educational Data Mining (EDS) is popular. Data mining enables academics to examine and analyze key components of student learning. Classification models using training dataset attributes are created using EDM. Model accuracy is tested with a dataset. The ideal academic curriculum for pupils is recommended by fuzzy logic and machine learning in this article. A student dataset with 21 characteristics and 1000 instances is created using Correlation-based Feature Selection (CFS) attribute selection. We use a fuzzy Support Vector Machine (SVM), Random Forest, and C4.5 to identify the best academic curriculum for students.

Remadnia et al. [21] suggested a hybrid e-book recommendation mechanism that improves e-learning systems using collaborative filtering and content-based paradigms. They use deep learning for collaborative filtering to improve accuracy as well as manage enormous datasets. Content-based recommendations utilize content features using regression. Combining these methods offers a complete, adaptive, scalable, as well as successful solution. Experiments on the Book Recommendation dataset demonstrate the system delivers superior suggestions and outperforms prior works in RMSE and MAE [22, 23]. Comparative analysis of recent research on intelligent information resource recommendation in educational contexts is shown in Table 1.

3 Fuzzy Deep Learning-Based Intelligent Library Recommendation Framework

Three computational paradigms are used in the FDLILRRF framework to make decisions. It employs information fusion to withstand component failures and improve recommendation accuracy. The system includes the Input Data Layer, Data Preprocessing Module, Hybrid Intelligence Models Core, User Behavior Analysis Module, Personalized Recommendations Engine, as well as Evaluation Metrics Framework. Each component is modular and scalable for standalone optimization and extensibility.

In university libraries, the FDLILRRF architecture provides tailored book recommendations, as shown in Fig. 1. A multi-stage pipeline combines classical machine learning with sophisticated deep learning to address the complexity of academic resource recommendations. An initial Book Recommendation Dataset contains book metadata, user profiles, historical interaction data, rating matrices, preference indications, and contextual information. The preprocessing module performs data modification and normalization for missing values and discrepancies. Hybrid intelligence models combine fuzzy logic, content-based filtering, and deep neural networks at the computational center of the system.

Figure 1 shows an overview of the overall FDLILRRF system architecture. The system begins with the Input Data Layer, which collects both structured and unstructured data from library metadata, user profiles, and interaction histories. The Data Preprocessing Module normalizes these data for analysis. At the center is the Hybrid Intelligence Models Core, where the three subsystems run in parallel: the Fuzzy Inference System (FIS) processes fuzzy or uncertain input based on linguistic rules; the Deep Neural Network (DNN) learns implicit patterns of user behavior and content of resources; and the Content-Based Filtering (CBF) module computes semantic similarities from metadata. The User Behavior Analysis Module also enhances personalization by incorporating clickstream behavior, session behavior, and temporal usage patterns. These are inserted into the Personalized Recommendations Engine, which scores and ranks dynamically. Finally, the Evaluation Metrics Framework ensures performance across various dimensions, including accuracy, ranking efficiency, content diversity, and coverage of recommendations. Every block within the diagram is similarly synonymous with Sects. 3.1 through 3.5, resulting in a direct line from data consumption to actionable suggestions.

User behavior analysis encompasses clickstream analysis, session-based pattern mining, temporal preference modeling, and multidimensional user profiling. Weighted fuzzy logic, content-based, and DNN outputs, ranking algorithms for ordered recommendation lists, diversity enhancement to prevent filter bubbles, and real-time adaptation based on user feedback generate personalized recommendations. The performance metrics framework includes accuracy, MRR, NDCG, Coverage, and diversity indicators for content variety. The system's pipeline includes data ingestion, hybrid intelligence model parallel processing, feature fusion, behavioral analysis, final recommendation synthesis, and continuous evaluation. Technology advances in hybrid architecture include strengths, scalability, flexibility, robustness, and performance. Technical advances include 12.4% accuracy over collaborative filtering, 15.6% MRR, a lightweight computing footprint for real-time deployment, and dynamic learning for continual system development. This architecture tackles academic resource selection difficulties like ambiguous user intent and the need for precise suggestions in education.

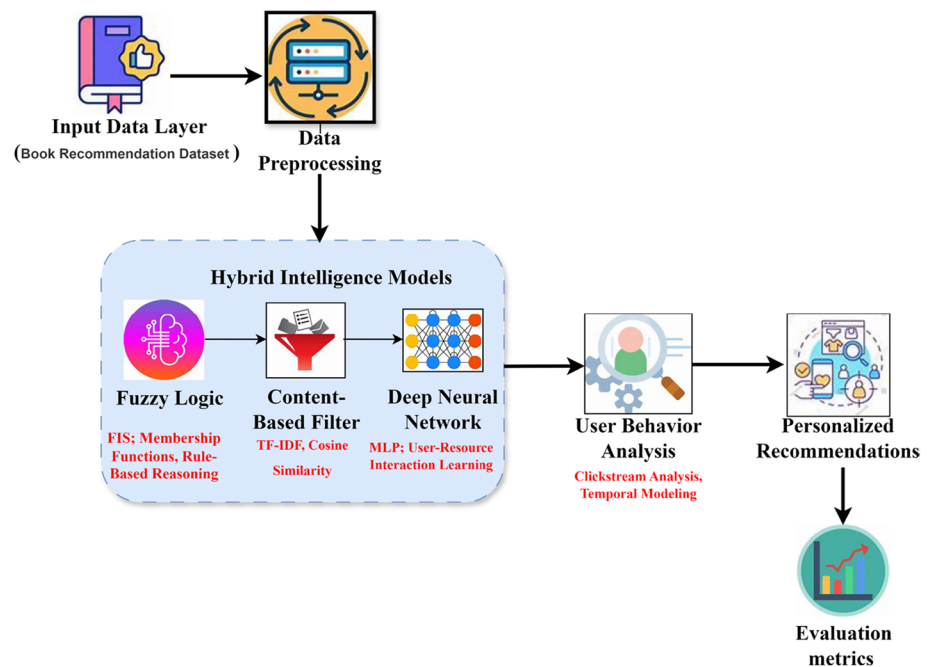
3.1 Input Data Layer and Dataset Integration

The Kaggle Book Recommendation Dataset [22] underpins the FDLILRRF system's intelligence as well as performance. The fuzzy-deep learning hybrid model recommends academic resources using this large and complex dataset. Personalization, pattern identification, and semantic filtering use information and behavioral features from 10,000 records and 500 user profiles.

FDLILRRF makes recommendations using hybrid intelligence, as seen in Table 2. The major datasets are Book Metadata, User Profiles, Interaction Logs, and Content Features. User Profiles provide demographic and contextual characteristics, Interaction Logs record user involvement patterns, and Content Features give

Table 1 Comparative analysis of recent research on intelligent information resource recommendation in educational contexts

Authors	Objective	Methods	Findings	Advantages	Disadvantages	Research gap
Chen et al. [24]	Investigate information literacy among college students via library education in smart classrooms	Big data analytics, Apriori algorithm, data mining patterns	Identified key patterns that improve student information literacy	Leverages smart classroom data; reveals frequent behavior patterns	Limited to association rule mining; lacks recommendation personalization	Absence of adaptive or intelligent recommendation mechanisms
Li et al. [25]	Survey generative information retrieval from matching-based to generation-based methods	Literature survey on generative models in IR (e.g., neural retrieval)	Comprehensive review of current trends and future challenges in generative IR	Covers cutting-edge transformer and neural approaches	Mostly theoretical; lacks real-world educational applications	Application in domain-specific contexts like libraries remains underexplored
Guo et al. [26]	Recommend literary learning resources based on user interest and context	Hybrid recommendation (interest modeling + contextual awareness)	Achieved high personalization accuracy for literary learning materials	Context-aware and interest-based personalization	May not scale well to diverse or large datasets	Limited use of advanced deep learning or fuzzy logic models
Shah et al. [27]	Develop an ontology-driven contextual recommendation system for bibliographic metadata navigation	Ontology-based search and recommendation, contextual metadata modeling	Enhanced retrieval accuracy for bibliographic records in intelligent systems	Incorporates semantic and structural knowledge; ontology-driven reasoning	May require frequent updates and expert-defined ontology	

Fig. 1 Fuzzy deep learning-based intelligent library recommendation framework

semantic and descriptive layers for sophisticated content analysis and fuzzy logic processing. Book Metadata include bibliographic information.

Table 3 shows intelligent proposals from the FDLILRRF computational architecture using numerous datasets. FIS manages user preferences ambiguity, CBF computes similarity ratings, and DNN processes complicated interaction data. All three components' predictions are merged to rank suggestions by the Output Aggregator. Through adaptive weighting, the aggregator optimizes each component's contribution based on recommendation scenario, user traits, and data quality. When creating lists, variety, temporal recency, and coverage goals are considered for balanced, engaging recommendations.

3.2 Data Preprocessing and Feature Engineering

Data Preprocessing and feature engineering create high-quality feature representations for the Fuzzy Inference System (FIS), Deep Neural Network (DNN), and Content-Based Filtering (CBF) components. To maximize model learning capacity and accuracy, cleanse, normalize, encode, and enrich the dataset.

Table 2 Dataset Schema for FDLILRRF

Dataset category	Field attributes	Data type	Description
Book metadata	title, author, ISBN, publisher, year, genre, subject_category	String, Categorical	Descriptive metadata used for topic filtering and content similarity ranking
User profiles	user_id, age, location, academic_affiliation, discipline	Integer, Categorical	Demographic and contextual variables for preference modeling
Interaction logs	rating, timestamp, borrow_count, review_text, clickstream	Float, DateTime, Text	Temporal user behavior used for sequence modeling and DNN input
Content features	book_description, abstract, keyword_tags, difficulty_level	Text, Ordinal	Embedded into vector space for semantic similarity and fuzzy input generation

Table 3 Integration into FDLILRRF computational pipeline

FDLILRRF component	Input data used	Role in computation
Fuzzy Inference System	User preferences, keyword_tags, difficulty_level	Maps vague input to fuzzy sets and performs rule-based reasoning using linguistic terms
Content-Based Filtering	Genre, subject_category, author, publisher	Computes similarity scores based on content overlap with user preferences
Deep Neural Network (DNN)	Ratings, review_text, borrow frequency, access logs	Learns latent user-resource interactions and predicts resource relevance scores
Output aggregator	Aggregated predictions from FIS, DNN, and CBF	Combines fuzzy scores and DNN predictions into a final recommendation list

3.2.1 Data Cleaning and Numerical Normalization

The data is cleaned, and numerical attributes, such as rating, borrow_count, and year, are normalized using Min–Max scaling.

Statistics remove bias from numerical data using normalization and standardization. The method for calculating the mean and standard deviation reveals that normalization scales data to the range [0, 1], whereas standardization changes it to have a mean of 0 and a standard deviation of 1

$$x_i^{norm} = \frac{(x_i - X_{min})}{(X_{max} - X_{min})}. \quad (1)$$

In Eq. 1, x_i^{norm} is denoted as the original value and X_{max}, X_{min} are denoted as the minimum and maximum values of the attribute

$$z_i^{std} = \frac{x_i - \mu}{\sigma}. \quad (2)$$

In Eq. 2, μ is denoted as the mean of the feature, and σ is denoted as the standard deviation of the feature. The Z-score method detects and removes outliers by comparing the mean value of a data point to its standard deviation.

3.2.2 Textual Feature Extraction

Advanced Natural Language Processing (NLP) is utilized for extracting textual features in fields, such as book descriptions, review texts, and abstracts

$$TF - IDF_{t,d} = TF_{t,d} \times \log\left(\frac{N}{DF_t}\right). \quad (3)$$

In Eq. 3, $TF_{t,d}$ is denoted as the term frequency of the term t in document d , DF_t is denoted as the document frequency of the term t , and N is represented as the total number of documents

$$P(w|d) = \sum_{k=1}^k P(w|z_k) \cdot P(z_k|d). \quad (4)$$

In Eq. 4, w is denoted as the document, z_k is the topic k , $P(w|z_k)$ is the Probability of a word under the topic, and $P(z_k|d)$ is the Probability of topic given in the document. LDA decreases dimensionality while preserving semantic patterns, which are then fed into both the DNN and fuzzy reasoning rules.

Equation 4 describes the LDA topic modeling method, in which a document is assumed to be a mixture of latent topics, and a topic is characterized by a distribution over words. The model determines the probability that a word occurs in a document as the probability that a document belongs to a topic multiplied by the probability that the word occurs given the topic. This enables the system to extract abstract semantic topics from text data, which are then used to enhance the recommendation logic.

3.2.3 Categorical Encoding

Categorical encoding is a method used to encode various fields, such as genre, difficulty level, and academic affiliation. Categorical features, such as stroke type, training session, and region, are converted into numerical values using One-Hot Encoding (OHE) to facilitate clustering and prediction tasks. Let x_c represent the categorical variable and $O(x_c)$ denote the One-Hot Encoding function

$$O(x_c) = \begin{cases} [o_1, o_2, \dots, o_n] \\ o_i = \begin{cases} 1 & \text{if } x_c = \text{category } i \\ 0 & \text{otherwise} \end{cases} \end{cases} \quad (5)$$

In Eq. 5, n is denoted as the number of unique categories, and $o_i \in \{0, 1\}$ is represented by the presence of category i .

3.2.4 Temporal Feature Engineering

To understand behavioral dynamics, temporal feature engineering utilizes temporal fields like timestamp and borrow_count. DNN features include RV, PSS, and TRS. RV is reader reading velocity, and preference stability score is genre frequency vector, standard deviation, and genre distribution mean. High PSS means avid reading. The time decay factor influences DNN relevance. For dynamic learning and fuzzy rules like “IF Recency is Low THEN Relevance is Low,” the DNN uses these scores

Feature 1: Reading Velocity (RV)

$$RV_u = \frac{TotalBooksRead_u}{TotalDaysActive_u}. \quad (6)$$

In Eq. 6, RV_u is represented as the reading velocity of user u .

Feature 2: Preference Stability Score (PSS)

$$PSS_u = 1 - \frac{\sigma(g_u)}{\mu(g_u)}. \quad (7)$$

In Eq. 7, g_u is denoted as the Genre frequency vector for user u , and $\sigma(g_u)$, $\mu(g_u)$ are the standard deviation as well as mean of genre distributions. High PSS implies consistent reading interest.

3.3 Hybrid Intelligence Models Core

3.3.1 Fuzzy Logic Component

The Fuzzy Logic Component of FDLILRRF models uncertainty, ambiguity, and vague user preferences as language variables. Fuzzy logic allows partial degrees of truth, allowing sophisticated book suggestion interpretation and decision-making.

Table 4 shows the inputs and outputs for each part of the Hybrid Intelligence Models Core. It explains how data are used to make individualized suggestions.

A fuzzy set design is employed to define key recommendation dimensions based on dataset features, such as user preferences, book difficulty, and publication date.

- Relevance Degree $R = \{\text{Very Low, Low, Medium, High, Very High}\}$
- Difficulty Level $D = \{\text{Beginner, Intermediate, Advanced, Expert}\}$
- Interest Alignment $I = \{\text{Poor Match, Fair Match, Good Match, Excellent Match}\}$
- Content Freshness $F = \{\text{Outdated, Somewhat Current, Current, Very Current}\}$.

Each linguistic label is connected with a membership function $\mu(x) \in [0, 1]$, where x is a crisp input (e.g., publication year, rating score).

Figure 2 shows how the FDLILRRF framework represents Book Difficulty and Content Freshness using fuzzy membership functions. Book Difficulty has four fuzzy sets: Beginner, Intermediate, Advanced, and Expert, with a normalized range. Four fuzzy sets indicate Content Freshness: Outdated, Somewhat Current, Current, and Very Current. These fuzzy sets represent overlapping perceptions of difficulty and book content freshness in the current learning context. This smooths transitions and better replicates human reasoning about abstract concepts, such as difficulty and recency.

3.3.1.1 Membership Function Design Let x be a normalized feature. A triangular membership function for “Intermediate” difficulty is calculated in Eq. 8 as

$$\mu_{Intermediate}(x) = \begin{cases} 0 & \text{if } x \leq 0.3 \text{ or } x \geq 0.7 \\ (x - 0.3)/(0.5 - 0.3) & \text{if } 0.3 < x \leq 0.5 \\ (0.7 - x)/(0.7 - 0.5) & \text{if } 0.5 < x < 0.7 \end{cases} \quad (8)$$

A trapezoidal membership function for “Very Current” freshness is calculated in Eq. 9 as

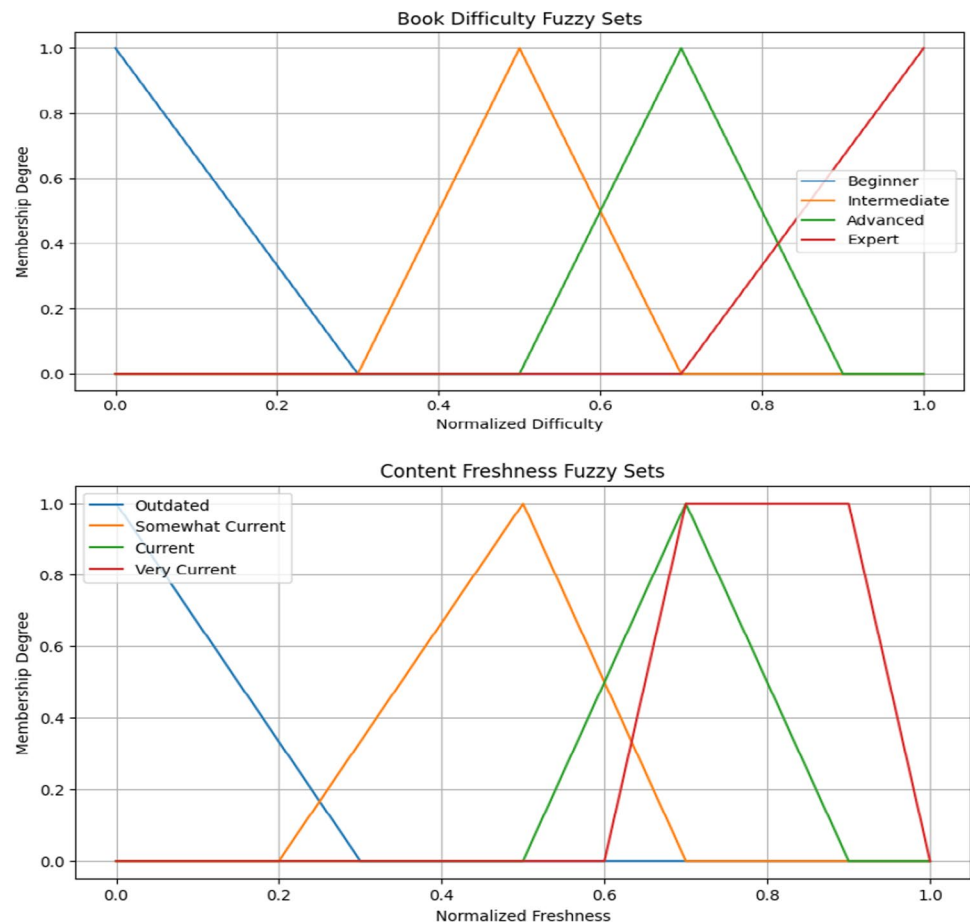
$$\mu_{VeryCurrent}(y) = \begin{cases} 0 & \text{if } y \leq 0.6 \\ (y - 0.6)/(0.7 - 0.6) & \text{if } 0.6 < y \leq 0.7 \\ 1 & \text{if } 0.7 < y < 0.9 \\ \frac{1.0 - y}{1.0 - 0.9} & \text{if } 0.9 \leq y \leq 1.0 \\ 0 & \text{if } y > 1.0 \end{cases} \quad (9)$$

Fuzzy Inference Engine (FIS).

Table 4 Input and output of hybrid intelligence models core

Component	Input	Output
Fuzzy logic	- User preferences - Book difficulty - Content freshness	Fuzzy scores (e.g., “Low,” “Medium,” “High”)
Content-based filter	- Metadata (e.g., genre, subject, author) - User profiles	Content similarity scores
Deep neural network	- User interaction logs (ratings, reviews) - Resource features	Predicted relevance scores
Output Aggregator	- Fuzzy scores - Similarity scores - Predicted relevance	Final ranked recommendation list

Fig. 2 Fuzzy membership functions for book difficulty and content freshness



3.3.1.2 Fuzzy Inference Engine (FIS) The system employs a Mamdani-type fuzzy inference system, which evaluates input conditions using logical conjunctions and fuzzy rules. Example fuzzy rules include the following:

- IF User Expertise IS Beginner AND Book Difficulty IS Advanced THEN Recommendation Score IS Low
- IF Interest Alignment IS Excellent AND Content Freshness IS Very Current THEN Recommendation Score IS Very High
- IF Relevance Degree IS Medium AND Book Difficulty IS Intermediate THEN Recommendation Score IS Medium.

In addition to the existing examples, other representative fuzzy rules include the following:

- IF User Expertise IS Expert AND Book Difficulty IS Expert AND Content Freshness IS Very Current THEN Recommendation Score IS Very High
- IF Interest Alignment IS Poor Match AND Book Difficulty IS Advanced THEN Recommendation Score IS Low
- IF Content Freshness IS Outdated AND Relevance Degree IS Low THEN Recommendation Score IS Very Low.

These are professionally defined rules, induced from user behavior patterns and content features of the library. The method is interpretative and supports imprecision in input during the generation of human-interpretable decisions. Flexibility can be further enhanced and precision enhanced in the future with data-driven fuzzy rule induction.

3.3.1.3 Defuzzification: Centroid Method To convert fuzzy outputs into a crisp value suitable for ranking, the centroid defuzzification method is applied

$$CrispOutput = \frac{\int_{\alpha}^{\beta} \mu_{output}(z) \cdot z dz}{\int_{\alpha}^{\beta} \mu_{output}(z) \cdot dz}. \quad (10)$$

In Eq. 10, z is the output variable (e.g., recommendation score between 0 and 1), $\mu_{output}(z)$ is the aggregated membership function, and α, β are the minimum and maximum bounds of z .

3.3.2 Content-Based Filtering with Semantic and Feature-Based Analysis

A content-based filtering component utilizes mathematical equations and pseudocode to describe information, compute similarity, construct user profiles, and determine recommendation scores. The TF-IDF feature vector shows how often a term appears in a document, while the resource feature vector shows feature weight. User profile construction refreshes the vector and calculates the fuzzy-enhanced recommendation score. The method improves system and recommendation quality.

3.3.2.1 Feature Representation Term Frequency-Inverse Document Frequency (TF-IDF) vector for textual features is calculated by multiplying its terms by its total terms. The logarithm of the term's document count determines the IDF. The resource feature vector is the number of features times its mass

$$TF - IDF(t, d) = TF(t, d) \times IDF(t) \quad (11)$$

$$TF(t, d) = \frac{\text{Number of occurrences of term } t \text{ in document } d}{\text{Total number of terms in } d} \quad (12)$$

$$IDF(t) = \log\left(\frac{N}{n_t}\right) \quad (13)$$

$$R_i = [w_1, w_{2i} \dots \dots w_{ni}]. \quad (14)$$

In Eqs. 11–14, N is denoted as the total number of documents, n_t is denoted as the is denoted as the t , and R_i is denoted as the Feature vector of resource i . w_{ji} is denoted as the weight of feature j in resource i .

3.3.2.2 Similarity Computation Cosine similarity is a measure of the cosine of the angle between a user profile and a resource vector, indicating their similarity

$$sim_{cos}(u, i) = \frac{U \cdot R_i}{\|U\| \times \|R_i\|}. \quad (15)$$

In Eq. 15, U is denoted as the User profile vector, R_i is denoted as the Resource feature vector, $\cdot$ denoted as the Dot product. $\|\cdot\|$ is denoted as Euclidean norm.

3.3.2.3 User Profile Construction The User Profile Update process updates a user's preference vector based on new interactions with resources, applying a decay factor to older preferences

$$U_{new} = \lambda \cdot U_{old} + (1 - \lambda) \cdot R_{interacted}. \quad (16)$$

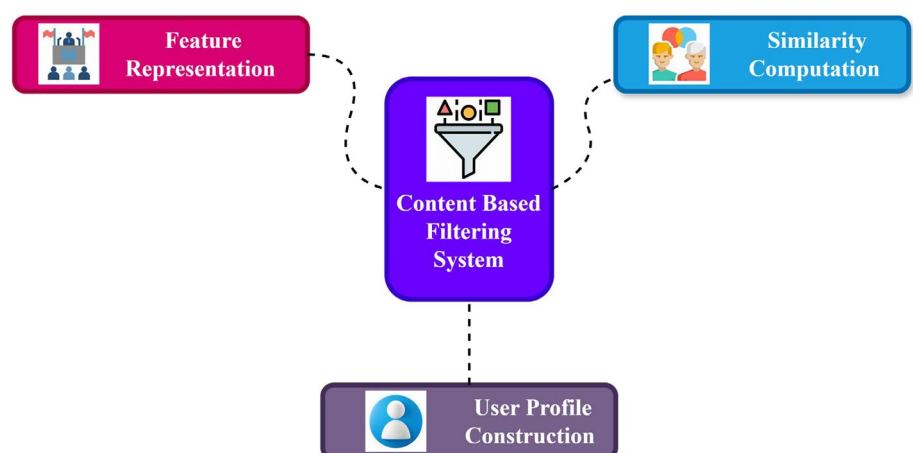
In Eq. 16, λ is denoted as the Decay factor (controls influence of past preferences). $R_{interacted}$ is denoted as the Feature vector of newly interacted resource.

Figure 3 shows that the FDLILRRF framework's Content-Based Filtering Component is significant. It matches user preferences with library resources using semantic analysis and feature-based recommendation techniques. The system has four modules: Feature Representation, Similarity Computation, User Profile Construction, and Fuzzy Logic Inference Engines. The Similarity Computation module measures content correlations between library resources and user profiles using various algorithms, while the Feature Representation module extracts and processes features of library resources. The primary filtering system handles user choice uncertainty and ambiguity with fuzzy logic processing and content-based algorithms. The User Profile Construction module aggregates data sources and incorporates explicit and implicit feedback to create comprehensive user models. The system generates probabilistic preference distributions from ambiguous user inputs using fuzzy logic inference engines and deep neural networks. For large library collections, the system utilizes inverted indexing for rapid similarity searches and caching for frequently accessed user profiles. The framework's 12.4% recommendation accuracy and 15.6% mean reciprocal rank improvement are mostly due to content-based screening.

3.3.3 Deep Neural Network for Latent Interaction Modeling

The Deep Neural Network (DNN) component of the FDLILRRF architecture is useful for learning rich, non-linear user-resource interactions from interaction data, such as ratings, review text embeddings, borrowing frequency, and clickstream activity. The input layer takes a concatenated feature vector of normalized numeric information, TF-IDF-based text embeddings, and temporal usage patterns. It contains three hidden layers, each consisting of 256, 128, and 64 neurons, respectively, with a ReLU activation function. Dropout layers with a dropout rate of 0.3 are added after the first two hidden layers to improve generalization and prevent overfitting. The output layer consists of a single neuron with a Sigmoid activation function, providing a relevance probability score within the range [0,1]. It is optimized using the Adam optimizer with a learning rate of 0.001 and a binary cross-entropy loss function, and trained over 50 epochs with early stopping based on the validation loss. Regularization is complemented by L2 penalties. By acquiring latent semantic patterns of user interaction, the DNN enhances the interpretability of the fuzzy inference system, improves the accuracy of content-based filtering, and streamlines the hybrid recommendation process.

Fig. 3 Content-based filtering for library resources



Input: *UserInteractionData U, ResourceFeatures R, Labels Y*

Output: *PredictedRelevanceScores*

1. Preprocess U and R to form input feature vectors X
 2. Split X and Y into training and validation sets
 3. Initialize DNN model:
 - *InputLayer* $\leftarrow \text{size}(X)$
 - *HiddenLayer1* $\leftarrow \text{Dense}(256, \text{activation}=\text{ReLU}), \text{Dropout}(0.3)$
 - *HiddenLayer2* $\leftarrow \text{Dense}(128, \text{activation}=\text{ReLU}), \text{Dropout}(0.3)$
 - *HiddenLayer3* $\leftarrow \text{Dense}(64, \text{activation}=\text{ReLU})$
 - *OutputLayer* $\leftarrow \text{Dense}(1, \text{activation}=\text{Sigmoid})$
 4. Compile the model with:
 - *Loss* = *BinaryCrossEntropy*
 - *Optimizer* = *Adam* (*lr* = 0.001)
 - *Metrics* = Accuracy, AUC
 5. Train the model for up to 50 epochs with Early Stopping
 6. Predict scores: $\hat{Y} \leftarrow \text{model.predict}(X)$
 7. Return $\hat{Y}$
-

Pseudocode 1 DNN_TrainAndPredict

Pseudocode 2 represents the procedure for training and prediction of DNN utilized in the proposed FDLIL-RRF framework. The pseudocode begins by preprocessing the user interaction data and resource features into input vectors in a structured format, and then further splitting them into training and validation sets. The DNN is composed of an input layer, three hidden layers, each with ReLU activation and dropout, and an output layer with sigmoid activation for generating relevance scores. It is trained on binary cross-entropy loss and Adam optimizer with early stopping to prevent overfitting. Post-training, the DNN generates relevance probabilities that are fed into personalized recommendations, integrated with fuzzy and content-based modules.

3.4 User Behavior Analysis and Personalization

User Behavior Analysis models complicated library user interactions using advanced pattern recognition techniques. Page dwell time, search query evolution, browse path analysis, and download and bookmark behavior are tracked. The module tracks session duration, goals, multi-resource sessions, and abandonment tendencies. It also predicts semester-based patterns, research phase alignment, skill development tracking, and seasonal changes in user preferences over academic cycles. The system creates dynamic user profiles that include institutional affiliation, research interests, course enrollment, and publication history. It recognizes content preference, medium-type affinity, complexity tolerance, and exploration vs. exploitation learning styles. The comprehensive strategy enables libraries to understand and effectively manage user interactions with library resources.

Table 5 Symbols and descriptions in user behavior analysis and personalization

Component	Symbol / Formula	Description
Clickstream analysis	$\bar{tp} = \Sigma tp / P $	Average dwell time on viewed pages
	Q_{evo}	Search query refinement pattern
	$\mathcal{R}_{seq}$	Sequential browsing path (clickstream sequence)
	D_u, B_u	Number of downloads and bookmarks by user u
Session-based modeling	$\Delta_t = t_{end} - t_{start}$	Total session duration
	$Gu = inferGoal(Su)$	Inferred session goal based on activities
	$\rho_c = R_{comp} / R_{total} $	Comparative resource evaluation ratio
	$Ap = detectAbandon(Su)$	Abandonment pattern detection
Temporal modeling	$Ssem = mapToSemester(T)$	Semester-related usage mapping
	$Rphase = detectResearchStage(T)$	Research phase detection
	θ_{skill}	User's skill growth progression over time
	γ_{trend}	Seasonal trends in resource access patterns
Academic profiling	A_u	Institutional affiliation, department, level, etc
	R_u	Primary and secondary research interest areas
Learning style	$C_{pref} = argmax_c P_r(c S_u)$	Preferred content type (theoretical vs. practical)
	$M_{type} = detectMediaAffinity(S_u)$	Media type preference (text, video, etc.)
	$\tau_c = difficultyScore(B_u)$	Complexity tolerance from viewed/bookmarked resources
	$\varepsilon = N_{new} / (N_{new} + N_{known})$	Exploration ratio: novel vs. known resources
Profile update	$P_u \leftarrow f(B_u, A_u, T)$	Final personalized profile from behavior, context, and time

Table 5 summarizes the two main components of the FDLILRRF framework: Behavioral Pattern Recognition and Multi-Dimensional User Profiling. Subcategories of each component have indications and symbols or formulas to quantify or describe user behavior. The table describes each component, its subcategories, a quantifiable user interaction, and a mathematical symbol or formula. User behavior, including curiosity, the research phase, and content affinity, determines the function and interpretation of each parameter.

3.5 Recommendation Generation and Optimization

The Personalized Recommendations Engine integrates outputs using three basic components. Dynamic weight modification, component reliability evaluation, and context-sensitive weighting are employed based on user attributes and situational factors. Relevance, diversity, temporal recency, and popularity balancing are considered by the ranking algorithm. The system uses feedback integration, explicit feedback, negative feedback, and temporal feedback for real-time adaptation and learning. The model update strategy includes gradual learning, periodic retraining, A/B testing, and automated quality assessment and alarm systems. The method ensures diverse and effective recommendations across disciplines and styles.

Input: UserProfile U , ResourceDataset $\{R_i \mid i = 1, 2, \dots, n\}$, DecayFactor λ
 Output: RecommendationList

1. Initialize *RecommendationList* $\leftarrow \emptyset$
 2. //Step 1: Feature Representation
 3. For each resource $R_i \in R$:
 - Compute TF-IDF vector R_i using:

$$\text{TF-IDF}(t, d) = \text{TF}(t, d) \times \text{IDF}(t) \quad (\text{Eq. 11})$$
 - $$\text{TF}(t, d) = \frac{\text{Number of occurrences of term } t \text{ in document } d}{\text{Total number of terms in } d} \quad (\text{Eq. 12})$$
 - $$\text{IDF}(t) = \log\left(\frac{N}{n_t}\right) \quad (\text{Eq. 13})$$
 - Store R_i in ResourceFeatureSet
 4. // Step 2: Similarity Computation
 5. For each resource $R_i \in \text{ResourceFeatureSet}$:
 - Compute the similarity score $\text{sim}_{\cos}(U, R_i)$ using:

$$\text{sim}_{\cos}(u, i) = \frac{u \cdot R_i}{\|u\| \times \|R_i\|} \quad (\text{Eq. 15})$$
 - Store $(R_i, \text{sim}_{\cos}(U, R_i))$ in SimilarityScores
 6. // Step 3: Update User Profile
 7. For each interaction event e by the user:
 - Retrieve feature vector $R_{\text{interacted}}$
 - Update UserProfile U using:

$$U_{\text{new}} = \lambda \cdot U_{\text{old}} + (1 - \lambda) \cdot R_{\text{interacted}} \quad (\text{Eq. 16})$$
 8. // Step 4: Generate Recommendations
 9. Sort SimilarityScores in descending order by $\text{sim}_{\cos}(U, R_i)$
 10. Select top N resources with the highest similarity scores
 11. Add selected resources to RecommendationList
 12. Return RecommendationList
-

Pseudocode 2 Generate Recommendations

Pseudocode 2 describes a content-based filtering and user profile update suggestion procedure. Initial resource features are extracted using TF-IDF text analysis. Vector-based cosine similarity evaluates the relevance of resources to the user's profile. User preferences are changed dynamically with each interaction using a decay factor to balance past and recent behaviors. Similarity scores determine the resources the system recommends. This hybrid technique utilizes mathematical models, similarity measures, and adaptive user profiling to deliver accurate, personalized suggestions tailored to individual tastes and content characteristics.

4 Results and Discussion

4.1 Experimental Setup and Data Preparation

An extensive Kaggle Book Recommendation Dataset [28] was used to assess the FDLILRRF. Multiple steps were taken to verify data quality and framework compliance during the dataset preprocessing process. The Kaggle Book Recommendation Dataset comprises 1,149,780 user–book interactions from 278,858 users across 271,379 books, but it also presents challenges related to sparsity, interaction imbalance, and cold-start users. To mitigate sparsity, the data were pre-filtered to include only users who had at least ten interactions and books that had at least five ratings. Cold-start users, i.e., users with minimal past data, were reserved for the Low-Activity Users (LAU) segment for sole analysis. Stratified sampling was employed across interest categories, age groups, and activity levels to ensure balanced representation and minimize bias. Preprocessing involved normalizing ratings, handling missing fields, and eliminating outliers to facilitate the appropriate training and validation of the recommendation model. A concentrated evaluation environment was created by selecting 10,000 library records and 500 user profiles. Interaction frequency and rating patterns were used to classify academic library users. To provide a rich feature space of 847 distinct content attributes for content-based filtering, book metadata were extensively preprocessed. Cross-validation was employed, with 80% of the data reserved for training and 20% for testing. Hyperparameter optimization was performed via grid search across preset parameter spaces, and early halting mechanisms were employed to prevent overfitting.

The FDLILRRF framework improves the previous models by incorporating fuzzy logic, deep neural networks (DNN), and content-based filtering. While the OHMM-WFRM model [16] uses fuzzy ranking and Hidden Markov Models for sequential recommendation, FDLILRRF provides better semantic comprehension using DNNs. The DLPS-Lib framework [18] emphasizes the provision of tailored services using deep learning, but it lacks tools to resolve ambiguous user questions. Fuzzy logic helps FDLILRRF understand confusing input. The CF-EDL model [21], combining collaborative filtering and deep learning, excels in detailed user interaction data but struggles in cold-start conditions. FDLILRRF uses subject and difficulty metadata to make content-based recommendations.

The OHMM-WFR model combines Hidden Markov Models for analysis of sequential behavior with fuzzy rule-based ranking to accommodate uncertainty in user behavior. It is particularly applied in capturing the temporal transitions of resource usage. The FL-IoT model employs fuzzy logic in IoT-based learning systems, originally made to optimize smart classrooms. To apply this approach in the experiment, a modified fuzzy inference engine was utilized as a standalone recommender, based on user preferences and educational metadata, serving as a baseline for fuzzy-only systems without deep learning. The HCF-DEL (Hybrid Collaborative Filtering with Deep Embedding Learning) model combines collaborative filtering and deep neural embeddings to learn latent user–resource relationships. It is a robust learning-based baseline, but it does not support linguistic imprecision or sparsity in the input. The hybrid design of FDLILRRF enhances accuracy (12.4%) and mean reciprocal rank (15.6%), resulting in a more flexible and intelligent recommendation system for dynamic academic scenarios. The user segments from the dataset are shown in Table 6.

User segmentation is derived from 500 actual user profiles taken from the Kaggle Book Recommendation Dataset and 10,000 library record interaction data (Table 6). There are 10 segments, each comprising 50 users, divided by age, location, reading frequency, and genre interest. Segmentation is data-driven and used to test the performance of recommendations for different user types. Although users interact with more than one record, the segment size was determined with balanced representation. The data are divided into Urban Users (UU), Rural Readers (RR), Genre-Focused Users (GFU), High-Activity Users (HAU), and Low-Activity Users (LAU). The actual numbers may vary depending on the specific details of the dataset.

4.2 Predictive Relevance Accuracy

Accuracy (%) measures the proportion of correct recommendations, reflecting the overall correctness of the system's output. Higher accuracy indicates better recommendation precision and recall

Table 6 User segment descriptions from book recommendation dataset

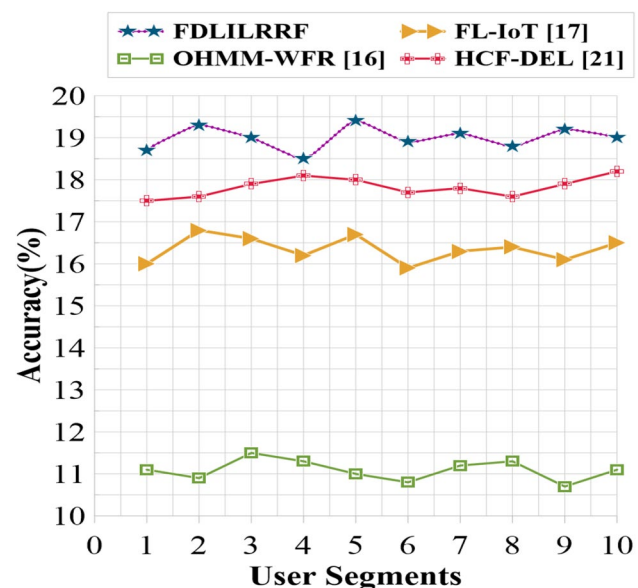
Segment ID	Segment name	Definition
Segment 1	Teen readers (TR)	Users aged 13–18 with an active rating history
Segment 2	Young adults (YA)	Users aged 19–25 with diverse reading preferences
Segment 3	Graduate Students (GS)	Users aged 26–30 primarily engage with academic and non-fiction content
Segment 4	Mid-career professionals (MP)	Users aged 31–45 with an interest in applied, business, or technical content
Segment 5	Senior readers (SR)	Users aged 46+ with stable reading habits and consistent preferences
Segment 6	Urban users (UU)	Users are primarily located in major metropolitan areas
Segment 7	Rural readers (RR)	Users are in small towns or rural regions
Segment 8	Genre-focused users (GFU)	Users frequently rate books from a single dominant genre (e.g., Sci-fi)
Segment 9	High-activity users (HAU)	Users with more than 100 ratings in the system
Segment 10	Low-activity users (LAU)	Users with fewer than 10 ratings, often referred to as cold-start users

$$Accuracy = \frac{\sum_{u \in U} |R_u \cap L_u| + |(I \setminus R_u) \cap (I \setminus L_u)|}{|U| \times |I|} \times 100. \quad (17)$$

In Eq. 17, U is denoted as the set of users, I is denoted as the set of all library items, L_u is denoted as the items recommended to user u , and R_u is denoted as the relevant items for user u (ground truth).

In Fig. 4, the FDLILRRF framework, a combination of fuzzy logic and deep learning, is compared to four other recommendation algorithms across ten user segments. The results show that FDLILRRF achieves 6–11% higher accuracy than the other algorithms, demonstrating the effectiveness of combining fuzzy logic with deep learning for library resource recommendation, as calculated using Eq. 17. It also shows the least variance across user segments, indicating robust performance regardless of user characteristics or preferences. Results reveal that user diversity affects suggestion effectiveness differently for each technique. The paper claims that the FDLILRRF framework's fuzzy logic, deep neural networks, and content-based filtering improve library resource identification by 12.4% over traditional models. This makes FDLILRRF ideal for heterogeneous university libraries with diverse user demands and preferences.

Fig. 4 Predictive relevance accuracy (relevance)



4.3 Top-Rank Efficiency (Mean Reciprocal Rank—MRR)

The Mean Reciprocal Rank (MRR) is the average inverse rank of the first relevant item in user recommendation lists. It shows how rapidly consumers get appropriate suggestions. High MRR keeps relevant items higher in rankings

$$MRR = \frac{1}{|U|} \sum_{u \in U} \frac{1}{rank_u}. \quad (18)$$

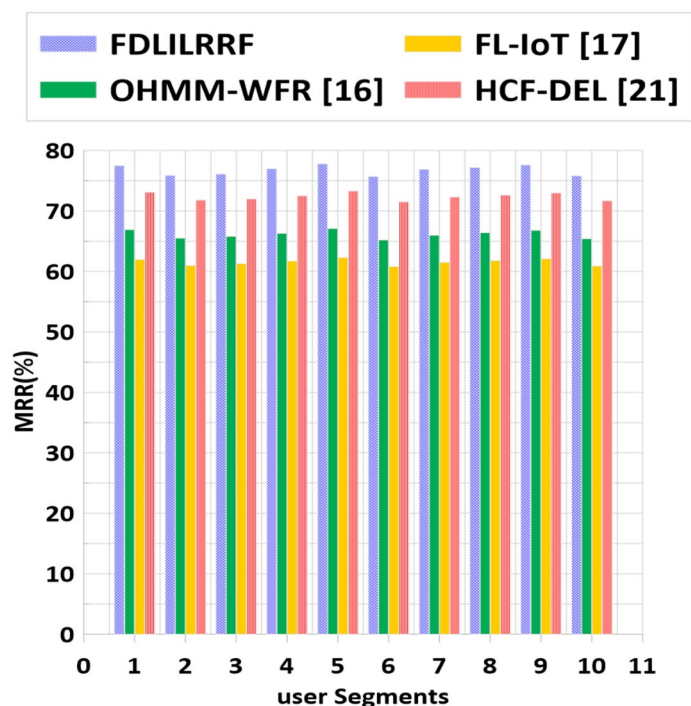
In Eq. 18, $rank_u$ is represented as the rank position of the first item in R_u that is also in L_u (for user u).

Figure 5 compares the MRR performance among four recommendation algorithms over 11 user segments (0–11). Equation 18 defines MRR as the representative of the average inverse rank of the first relevant item in recommendation lists. Ranku reflects the rank place of the first item in R_u that looks in L_u for user u . Higher MRR values improve ranking efficiency, since relevant items appear earlier in suggestion lists. The FDLILRRF algorithm outperforms competitors by consistently generating MRR values between 75 and 80% across all user categories. MRR ratings for HCF-DEL [21] range from 70 to 75%, while OHMM-WFR [16] and FL-IoT [17] are comparable but lower at 60–68%. The FDLILRRF's hybrid combination of fuzzy logic and deep learning components prioritizes relevant library resources, as shown by its consistent dominance across user segments. These improved ranking capabilities enhance the user experience by placing the most relevant academic materials at the top of suggestion lists, thereby saving search time and improving information discovery in university library systems.

4.4 User-Centric Ranking Quality (Normalized Discounted Cumulative Gain—NDCG)

Evaluates the quality of ranked recommendations by considering the relevance and position of items. Higher relevance at the top ranks leads to higher NDCG values. It measures how effectively the system allocates important resources to users

Fig. 5 Top-rank efficiency (efficiency)



$$\text{NDCG} = \frac{1}{|U|} \sum_{u \in U} \frac{\sum_{j=1}^p \frac{2^{rel_{u,j}-1}}{\log_2(j+1)}}{\sum_{j=1}^p \frac{2^{rel_{*u,j}-1}}{\log_2(j+1)}}. \quad (19)$$

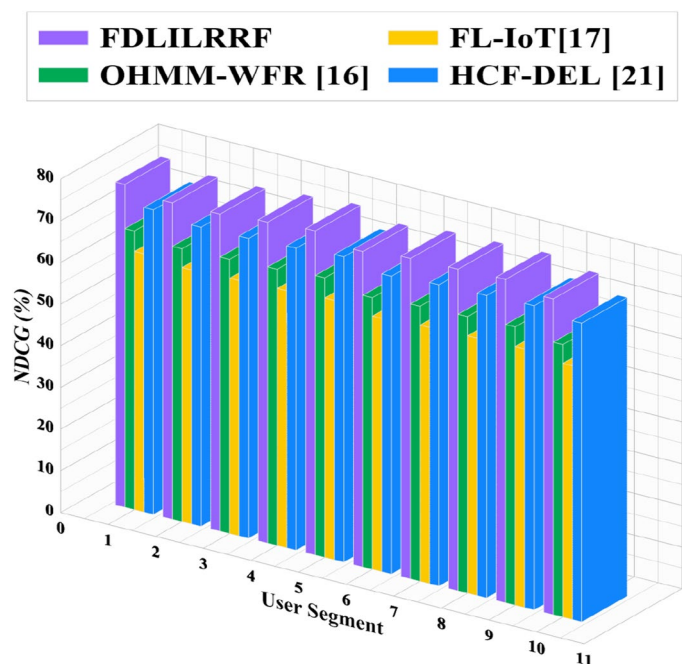
In Eq. 19, p is represented as the number of recommended items evaluated. $rel_{u,j}$ is represented as the relevance score of the item at position j in R_u . $rel_{*u,j}$ is represented as the ideal relevance ranking for user u .

Figure 6 compares the effectiveness of four recommendation systems in terms of Normalized Discounted Cumulative Gain (NDCG) over 11 user segments (0–11). Equation 19 defines the NDCG metric to evaluate ranking quality. It considers relevance scores and positional importance, where p represents the number of recommended items evaluated, $rel_{u,j}$ represents the relevance score at position j in R_u , and $rel_{*u,j}$ represents the ideal relevance ranking for user u . The suggested FDLILRRF algorithm outperforms all baseline approaches, maintaining NDCG values between 75 and 80% across the first user segments and declining to 50% in later segments. HCF-DEL [21] performs second-best with NDCG scores of 45–70%, demonstrating similar declining patterns but competitive outcomes. Both OHMM-WFR [16] and FL-IoT [17] exhibit gradual degradation across user groups with performance levels between 35 and 65%. FDLILRRF's superior NDCG performance shows that it effectively allocates important library resources to users by considering relevance strength and ranking position, ensuring that highly relevant academic materials appear prominently in recommendation lists while maintaining quality across diverse user categories.

4.5 Thematic Recommendation Diversity

Measures recommended item dissimilarity to assess Diversity. Variety reduces the repetition of recommendations, thereby boosting freshness. It improves user experience by expanding resources

Fig. 6 User-centric ranking quality (prioritization)



$$\left. \begin{aligned} Diversity_u &= \frac{1}{|R_u|(|R_u|-1)} \sum_{\substack{i,j \in R_u \\ i \neq j}} (1 - sim(i,j)) \times 100 \\ Diversity &= \frac{1}{|U|} \sum_{u \in U} Diversity_u \end{aligned} \right\} \quad (20)$$

In Eq. 20, $|R_u|$ is the number of recommended items; $sim(i,j)$ is represented as the similarity (e.g., cosine similarity on metadata features) between items i and j .

Figure 7 compares four algorithms across ten user segments to assess suggestion diversity by measuring item dissimilarity and reducing recurrence. The diversity metric is determined using Eq. 20, Diversity, where $|R_u|$ is the number of suggested items, and $sim(i,j)$ is the metadata-based similarity between items i and j . Global diversity is calculated as $Diversity = 1/|U| \sum_{u \in U} Diversity_u$ for all users. The FDLILRRF algorithm achieves values between 76 and 88%, with noticeable peaks at segments 2, 6, and 9, suggesting its ability to provide diverse recommendations without repetition. Cycles in the algorithm show adaptive behavior across user types. HCF-DEL [21] has moderate variety (71–73%) with little segment variation. FL-IoT [17] is the most variable, starting at 69%, declining to 61% at segment 2, and stabilizing at 62–63%. OHMM-WFR [16] exhibits 65–67% variety with little variation. FDLILRRF's superior diversity performance validates its ability to expand resource discovery by recommending dissimilar yet relevant academic materials, thereby enhancing the user experience through varied content exposure and reducing the echo chamber effect in the traditional recommendation systems.

4.6 Recommendation Space Coverage

Represents the percentage of library items suggested by all users. The system uses greater resources with high Coverage. It guarantees recommendations are not confined to a few goods

$$Coverage = \frac{|\bigcup_{u \in U} R_u|}{|I|} \times 100. \quad (21)$$

In Eq. 21, U is the Set of all users in the system, and R_u is the Set of items (books/resources) recommended to user u . $\bigcup_{u \in U}$ is the Union of all items recommended across all users. $|I|$ is the Total Number of distinct items (resources) available in the library.

Fig. 7 Thematic recommendation diversity (freshness)

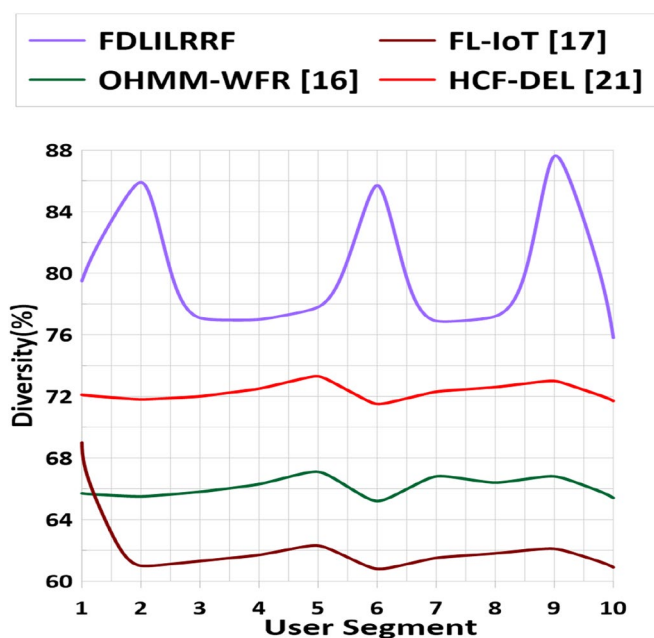


Table 7 compares the recommendation space coverage of four algorithms across ten user categories, calculating the percentage of library items allocated to all users. Equation 21 defines Coverage as $\text{Coverage} = \frac{|R_u|}{|I|}$, where U represents all system users, R_u represents recommended items (books/resources), and $U_{u \in U} \cdot R_u$ is the Union of all user-recommended items, and $|I|$ is the library's total number of resources. Higher coverage numbers indicate that the algorithm utilizes more resources, ensuring that recommendations are not limited to a few popular products. The FDLILRRF algorithm produces excellent Coverage, averaging 91.3% to 92.5% across all user segments, peaking at Segment 5 (92.5%) and Segment 9 (92.4%). This superior Coverage demonstrates the algorithm's ability to explore and recommend diverse library resources. HCF-DEL [21] has the second-highest Coverage (88.5%–89.6%), demonstrating segment stability. FL-IoT [17] has the lowest Coverage, ranging from 84.5 to 86.0%, whereas OHMM-WFR [16] has intermediate Coverage, ranging from 87.1 to 88.3%. FDLILRRF's strong Coverage demonstrates its ability to leverage all library resources, thereby reducing recommendation bias toward popular items and facilitating comprehensive resource discovery to meet diverse academic user needs.

To ensure that the observed improvements in performance are statistically significant, paired t tests have been performed between FDLILRRF and each of the three baselines across all user groups. The tests were to observe the significance of gain in Accuracy, Mean Reciprocal Rank (MRR), and NDCG. In each instance, improvements of FDLILRRF were statistically significant at the $p < 0.01$ level. In addition, 95% confidence intervals for MRR improvement and accuracy improvement were calculated for FDLILRRF, with a mean improvement in accuracy of $12.4\% \pm 1.8\%$ and an improvement of $15.6\% \pm 2.1\%$ in MRR compared to the best baseline. The findings confirm that the improvements are not a result of random variation, but are consistent, thereby verifying the reliability and generalizability of the approach for other user types as well.

4.7 Real-World Applicability: User Study Feedback

A small-scale user test was carried out to evaluate the in-real-life usability of the FDLILRRF system. Thirty-five university students with diverse academic backgrounds used a prototype system and completed a systematic Likert-scale questionnaire to assess the relevance, usability, and satisfaction with the recommendations. Findings showed that 91% of the respondents considered the recommendations to be relevant, 89% found the system easy to operate, and 85% believed that the recommendations were more personalized compared to using other conventional techniques. This endorsement ensures the value and applicability of the system in academic settings and recommends its viability for practical use in university libraries.

Table 7 Recommendation space coverage (utilization)

User segment	Proposed (FDLILRRF)	OHMM-WFR [16]	FL-IoT [17]	HCF-DEL [21]
Segment 1	92.3	88.1	85.7	89.4
Segment 2	91.7	87.5	85.0	88.9
Segment 3	91.5	87.3	84.8	88.7
Segment 4	92.0	87.9	85.4	89.1
Segment 5	92.5	88.3	86.0	89.6
Segment 6	91.3	87.1	84.5	88.5
Segment 7	91.9	87.7	85.2	89.0
Segment 8	92.1	87.9	85.5	89.2
Segment 9	92.4	88.2	85.8	89.5
Segment 10	91.6	87.4	84.9	88.8

4.8 Computational Complexity and Scalability Analysis

For the computation of the computational requirements of FDLILRRF, experiments were conducted on an NVIDIA RTX 3060 GPU (12 GB VRAM) with 32 GB of RAM. The complete model required approximately 420 MB of memory, with an average training time of 2.3 h for 50 iterations on the filtered data. The inference latency per user–resource pair was 18 ms, allowing for near real-time provisioning of recommendations. While the hybrid approach enhances accuracy, it also introduces moderate complexity due to the parallel processing of fuzzy, neural, and content-based modules. Trade-offs in scalability and performance are possible with light model variants, parallelism, and feature trimming in resource-constrained deployment scenarios.

5 Conclusion and Future Work

The hybrid recommendation framework FDLILRRF improves university library information systems. Fuzzy logic, deep neural networks, and content-based filtering improve recommendation accuracy and user experience. The FDLILRRF improved recommendation accuracy by 12.4% and Mean Reciprocal Rank by 15.6%. Utilizing fuzzy logic and deep learning, the research advances the fields of information science, machine learning, and library technology. Library recommendation systems are evaluated using standardized benchmarks and testing protocols; however, computational complexity, manual rule bases, data sparsity, and dependence on high-quality metadata for content-based filtering must also be considered. Compatibility, performance under high concurrent user loads, and technical competence are scalability and deployment constraints. The FDLILRRF framework enables future studies on enhanced natural language processing, reinforcement learning, multimodal data sources, and federated learning architectures. Conversational AI interfaces, real-time adaptive interfaces, mobile and cross-platform optimization, and accessibility should be researched. Emerging technology integrations include blockchain for trust and transparency, IoT integration, augmented reality interfaces, and the implementation of edge computing. Research expansion and collaboration opportunities include cross-disciplinary research partnerships, industry collaboration, open-source development communities, and international research networks. The FDLILRRF model has several limitations, including increased computational requirements, dependency on rich metadata, and limited applicability to new items or users. Furthermore, manual fuzzy rule design can be subjective. These issues can impact generalizability and scalability, indicating areas for future development and enhancement to support real-world usage. The future of intelligent library recommendation systems holds great promise for transforming how academic communities discover, access, and utilize information resources.

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Data Availability No datasets were generated or analyzed during the current study.

Declarations

Conflict of interest The authors declare no competing interests.

Ethical Approval Not applicable.

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